

Summary report

Wellbore: 15/9-F-15 D

Period: 2013-11-30 00:00 - 2013-12-01 00:00

Status:	normal
Report creation time:	2018-05-03 13:52
Report number:	21
Days Ahead/Behind (+/-):	13.7
Operator:	Statoil
Rig Name:	MÆRSK INSPIRER
Drilling contractor:	Maersk Drilling
Spud Date:	2013-11-10 14:00
Wellbore type:	
Elevation RKB-MSL (m):	54.9
Water depth MSL (m):	91
Tight well:	Y
HPHT:	Y
Temperature (I):	
Pressure (I):	
Date Well Complete:	2013-12-02

Dist Drilled (m):	522
Penetration rate (m/h):	-999.99
Hole Dia (in):	8.5
Pressure Test Type:	formation integrity test
Formation strength (g/cm3):	1.6
Dia Last Casing (I):	

Depth at Kick Off mMD:	
Depth at Kick Off mTVD:	
Depth mMd:	4434
Depth mTVD:	3169
Plug Back Depth mMD:	
Depth at formation strength mMD:	3302
Depth At Formation Strength mTVD:	2848
Depth At Last Casing mMD:	3294.7
Depth At Last Casing mTVD:	

Summary of activities (24 Hours)

Drilling and geosteering 8 1/2" hole from 3912 m to 4434 m.

Summary of planned activities (24 Hours)

Drilling and geosteering 8 1/2" hole to TD. Circulate well clean at TD. Take torque readings. POOH with 8 1/2" BHA to 9 5/8" shoe. Re-log 1 stand. Take torque readings. POOH with 8 1/2" BHA.

Operations

Start time	End time	End Depth mMD	Main - Sub Activity	State	Remark
00:00	01:45	3954	drilling -- drill	ok	Drilled 8 1/2" hole from 3912 m to 3954 m. with 2400 lpm, 219 bar, 120 rpm, 17-25 kNm, 5 ton WOB. Avg ROP 24 m/hr. ECD 1,60 SG. Max gas: 2,95 %. Down link and take survey after connection.
01:45	02:30	3954	drilling -- other	ok	At 3954 m observed STDP increased from 220 bar to 245 bar and WOB increased from 4 ton to 9 ton. Drillstring stalled out. Attempted to regain rotating no success. Pulled 70 ton overpull on drillstring, string free. Back reamed 1 single prior to commence drilling.
02:30	03:45	3974	drilling -- drill	ok	Drilled 8 1/2" hole from 3954 m to 3974 m. with 2400 lpm, 220 bar, 120 rpm, 18-23 kNm, 4 ton WOB. Avg ROP 16 m/hr. ECD 1,51 SG. Max gas: 0,8 %. Down link and take survey after connection. Back reamed stand prior to connection.
03:45	06:00	4025	drilling -- drill	ok	Drilled 8 1/2" hole from 3974 m to 4025 m. with 2400 lpm, 220 bar, 120 rpm, 18-23 kNm, 4-5 ton WOB. Avg ROP 22 m/hr. ECD 1,53 SG. Max gas: 3,0 %. Down link and take survey after connection.
06:00	07:15	4052	drilling -- drill	ok	Drilled 8 1/2" hole from 4025 m to 4052 m. with 2400 lpm, 226 bar, 120 rpm, 22-26 kNm, 5 ton WOB. Avg ROP 22 m/hr. ECD 1,53 SG. Max gas: 2,4 %. Down link and take survey after connection.
07:15	07:30	4052	drilling -- other	ok	Took SCR @ 4052 m, 1,32 SG. Pump 1. 20/30/40 stroke, 33/41/54 bar. Pump 2. 20/30/40 stroke, 34/40/51 bar.
07:30	20:00	4337	drilling -- drill	ok	Drilled 8 1/2" hole from 4052 m to 4337 m. with 2400 lpm, 238 bar, 140- 160 rpm, 20-28 kNm, 4-6 ton Avg WOB. ROP 23 m/hr. ECD 1,56 SG. Max gas: 2,9 %. Down link and take survey after connection.
20:00	20:15	4337	drilling -- other	ok	Took SCR @ 4337 m, 1,32 SG. Pump 1. 20/30/40 stroke, 34/43/55 bar. Pump 2. 20/30/40 stroke, 36/43/53 bar.
20:15	00:00	4434	drilling -- drill	ok	Drilled 8 1/2" hole from 4337 m to 4434 m. with 2400 lpm, 236 bar, 160 rpm, 24-30 kNm, 4-5 ton WOB. Avg ROP 26 m/hr. ECD 1,56 SG. Max gas: 2,6 %. Down link and take survey after connection.

Drilling Fluid

Sample Time	04:15	09:00	14:00	20:30
Sample Point	Flowline	Flowline	Flowline	Flowline
Sample Depth mMD	4525	4090	4201	4338
Fluid Type	Enviromul Yellow	Enviromul Yellow	Enviromul Yellow	Enviromul Yellow
Fluid Density (g/cm3)	1.32	1.32	1.32	1.32
Funnel Visc (s)	-999.99	-999.99	-999.99	-999.99
Mf ()				
Pm ()				
Pm filtrate ()				
Chloride ()				
Calcium ()				
Magnesium ()				
Ph				
Excess Lime ()				
Solids				
Sand ()				
Water ()				
Oil ()				
Solids ()				
Corrected solids ()				
High gravity solids ()				
Low gravity solids ()				
Viscometer tests				
Plastic visc. (mPa.s)	33	38	36	34
Yield point (Pa)	13	13	12	10.5
Filtration tests				
Pm filtrate ()				
Filtrate Lthp ()				

Filtrate Hthp ()				
Cake thickn API ()				
Cake thickn HPHT ()				
Test Temp HPHT (degC)	120	120	120	120
Comment				

Pore Pressure

Time	Depth mMD	Depth TVD	Equ Mud Weight (g/cm3)	Reading
00:00	3300		1.14	estimated
00:00	3449		1.29	estimated
00:00	4434		1.14	estimated

Survey Station

Depth mMD	Depth mTVD	Inclination (dega)	Azimuth (dega)	Comment
3959.1	3055.2	68.18	111.6	
3999.8	3069.3	71.27	111.64	
4041.1	3081.3	74.92	112.17	
4081.4	3091.8	74.89	111.29	
4121.4	3102.2	75.02	109.65	
4161.6	3112	76.92	109.69	
4174.2	3114.8	77.12	109.31	
4202.3	3120.6	78.85	108.76	
4242.1	3128.3	78.99	107.19	
4282.7	3136.1	78.9	106.62	
4322.7	3143.7	78.97	106.47	
4362.4	3151.8	77.5	106.91	
4403.7	3161.3	75.94	107.23	
4443.4	3170	78.75	107.82	

Log Information

Run No	Service Company	Depth Top mMD	Depth Bottom mTVD	Tool	BHST (degC)
105	Baker Hughes Inteq	3912	-999.99	ZoneTrak G - CoPilot - OnTrak - ORD - C C	-999.99

Lithology Information

Start Depth mMD	End Depth mMD	Start Depth mTVD	End Depth mTVD	Shows Description	Lithology Description
3912	4434	3036.8	3169		Sandstone with some claystone interbeds and minor Limestone stringers

Gas Reading Information

Time	Class	Depth to Top mMD	Depth to Bottom MD	Depth to Top mTVD	Depth to Bottom TVD	Highest Gas (%)	Lowest Gas ()	C1 (ppm)	C2 (ppm)	C3 (ppm)	IC4 (ppm)	IC5 (ppm)
00:00	drilling gas peak	3911		3036		3.43		31589	2924	1507	142	100
00:00	drilling gas peak	4033		3078.7		3.15		29223	2682	1293	114	76
00:00	drilling gas peak	4131		3104.6		3.06		28550	2639	1291	113	73
00:00	drilling gas peak	4148		3108.4		2.9		26089	2499	1274	115	75
00:00	drilling gas peak	4174		3114.5		3.08		27279	2664	1400	129	86
00:00	drilling gas peak	4193		3118.5		2.98		26303	2599	1395	131	87
00:00	drilling gas peak	4212		3122.5		2.88		25977	2360	1234	117	81
00:00	drilling gas peak	4235		3126.9		2.49		20485	2118	1195	118	84
00:00	drilling gas peak	4251		3129.8		2.58		21419	2195	1228	120	86
00:00	drilling gas peak	4283		3136.1		2.73		23221	2380	1302	125	89
00:00	drilling gas peak	4409		3162.4		2.63		22685	2263	1206	114	84